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# Nondisjunction and unequal spindle organization accompany the drive of *Aegilops speltoides* B chromosomes

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## Summary

- B chromosomes (Bs) are supernumerary chromosomes, which are often preferentially inherited. When transmission rates of chromosomes are higher than 0.5, not obeying the Mendelian law of equal segregation, the resulting transmission advantage is collectively referred to as 'chromosome drive'.
- Here we analysed the drive mechanism of *Aegilops speltoides* Bs.
- The repeat AesTR-183 of *A. speltoides* Bs, which also can be detected on the Bs of *Aegilops mutica* and rye, was used to track Bs during pollen development. Nondisjunction of CENH3-positive, tubulin interacting B sister chromatids and an asymmetric spindle during first pollen grain mitosis are key for the accumulation process. A quantitative flow cytometric approach revealed that, independent of the number of Bs present in the mother plant, Bs accumulate in the generative nuclei to > 93%. Nine out of 11 tested (peri)centromeric repeats were shared by A and B chromosomes.
- Our findings provide new insights into the process of chromosome drive. Quantitative flow cytometry is a useful and reliable method to study the drive frequency of Bs. Nondisjunction and unequal spindle organization accompany during first pollen mitosis the drive of *A. speltoides* Bs. The prerequisites for the drive process seems to be common in Poaceae.

## Introduction

Supernumerary B chromosomes (Bs) are considered to be dispensable chromosomes, which under standard conditions do not confer advantages on the organisms that harbour them. Bs are found in thousands of fungi, plant and animal species and are thought to stand for a specific type of selfish DNA (Jones & Rees, 1982). Recent advances in deciphering the sequence composition of B chromosomes provided a unique opportunity to analyse the origin and evolution of this enigmatic genome component. It was shown that the Bs of several plant and animal species harbour gene-derived sequences. This allowed their origin to be traced from duplicated fragments of multiple A chromosomes of their host species (reviewed in Ruban *et al.*, 2017). Hence, Bs could be considered as a by-product of standard chromosome (As) evolution (Martis *et al.*, 2012).

Often the transmission rate of B chromosomes is > 0.5, not obeying the Mendelian law of equal segregation. The resulting transmission advantage is collectively referred to as 'drive'. The maximum number of Bs tolerated by the host varies between species and depends on a balance between the B chromosome drive and the negative effects, especially on fertility and vigour, caused by B chromosomes if they occur in higher numbers (Bougourd & Jones, 1997). Among the relatively small number of B chromosome-positive species that have been investigated, relative to the total number of known species, there is a variety of mechanisms of B chromosome drive that occur before, during or after meiosis, and in some cases drive has not been found (reviewed in Carlson, 1988; Jones, 1991; Burt & Trivers, 2006; Houben, 2017; Jones, 2018). Besides drive, the nonMendelian inheritance of Bs could also be affected by mitotic and meiotic instabilities.

In *Aegilops speltoides* Tausch (syn. *Triticum speltoides* (Tausch) Gren.), an annual diploid Poaceae (genome type S), besides the seven pairs of A chromosomes, plants with a maximum number of eight additional Bs were reported (Raskina *et al.*, 2011). The particular number of Bs is constant in all aerial parts within the same individual. Interestingly, B chromosomes are completely absent in the roots (Mendelson & Zohary, 1972; Ruban *et al.*, 2014). Reciprocal crosses between 0B and +B plants indicated that the drive of Bs occurs during the first mitosis in the male gametophyte whereas there is no accumulation of Bs during the development of female gametophytes (Mendelson & Zohary, 1972). The nonMendelian transmission of Bs cannot be explained solely by irregular segregation during meiosis. The net increase of B number in the progeny of 0B × 1B crosses can only be accounted for by the directed nondisjunction of the Bs to the generative nucleus during first pollen mitosis as in rye.



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Here we analyse the drive of *A. speltoides* B chromosomes and provide direct insights into its cellular mechanism. We first employed comparative genomics to identify B-specific repeats, which we used to track the B chromosomes during microgametogenesis. It was found that sister chromatid nondisjunction and an asymmetric spindle during first pollen grain mitosis are components of the B chromosome accumulation process. To determine the B accumulation rate in sperm nuclei, we developed a novel flow cytometric approach that allowed the differentiation between vegetative and sperm nuclei as well as the quantification of B chromosomes inside the nuclei. Utilizing this method, we found that, independent of the number of B chromosomes present in the mother plant, Bs accumulate in the generative nuclei to > 93%. The prerequisites for the drive process seem to be common in Poaceae, thus enabling independent origins of Bs in different lineages within the family.

## Material and Methods

### Plant materials and cultivation

The following accessions were used: *Aegilops speltoides* from Syria (PI 487238, USDA-ARS, USA) and Israel (Katzir, TS 89; Technion, 2-36; Ramat Hanadiv, 2-46, Institute of Evolution, University of Haifa, Israel) (Belyayev & Raskina, 2013); *Aegilops mutica* (KU-12008, Plant Germplasm Institute, Kyoto University, Japan) (Ohta & Saruhashi, 1999); *Triticum aestivum* 'Chinese Spring' (TRI 12922, IPK-Gatersleben Genebank, Germany); *Secale cereale* inbred line 7415 (Jimenez *et al.*, 1994) and *Lolium perenne* accession G7 with Bs (Taylor & Evans, 1977). Plants were cultivated under glasshouse conditions (16 h 22°C : 8 h 16°C light : dark).

### Isolation and sequencing of genomic DNA

Genomic DNA from leaf tissue was purified with the DNeasy Plant Mini Kit (Qiagen). Using the Illumina HiSeq 2500 system, we sequenced 2 × 101 bp genomic DNA from 0B and 2B plants from the *A. speltoides* Katzir accession. In total, 171 million Illumina 101 bp paired-end reads were generated having an output of residues of 17 Gbp. All sequences are publicly available under PRJEB29862 (ENA).

### Identification of B-specific repeats using similarity-based read clustering

Identification of *A. speltoides* B chromosome-specific repeats was performed by similarity-based clustering of Illumina reads as implemented in the RepeatExplorer pipeline (Novák *et al.*, 2010, 2013). The analysis was done in a comparative mode, using as its input a randomly sampled fraction of 1.4 million paired-end reads from 0B and +B samples, corresponding to 0.03 × genome coverage. Following the clustering of the pooled reads and annotation of repeats represented by the largest clusters containing at least 0.01% of analysed reads, the proportions of reads in each cluster from 0B and +B samples were determined by parsing read names where genotypes of origin were encoded by specific tags. This allowed us to identify repeats enriched on B chromosomes as the clusters with increased proportions of reads from the +B sample compared with the 0B sample. Monomer lengths and consensus sequences of newly identified satellite repeats were determined using Tarean (Novák *et al.*, 2017). PCR primers were designed for consensus sequences of identified B-repeats (Supporting information Tables S1, S2). Control primers were designed for the *T. aestivum* 18S ribosomal RNA coding part of rDNA 25S-18S intergenic region *Eco*RI-*Bam*HI fragment (GenBank accession X07841.1). 0B/+B genomic DNA was used as a template and PCR products were cloned using the CloneJET PCR Cloning Kit (Thermo Fisher Scientific, Waltham, MA, USA) and verified by sequencing (GenBank nos. AesTR-126 MK283665, AesTR-148 MK283666, AesTR-183 MK283667, AesTR-205 MK283668).

### Molecular phylogenetic analysis of AesTR-183-like repeats

AesTR-183-like sequences were amplified by PCR from genomic DNA of *A. mutica*, *S. cereale* and three *A. speltoides* accessions (Katzir, Technion, Ramat Hanadiv), all containing B chromosomes. Sequences were analysed and assembled by Sequencher v.5.4.6 (Gene Codes Corp., Ann Arbor, MI, USA). These sequences, including a similar region of wheat chromosome 1B (accession LS992081.1, position 432930807–432930254), were aligned with ClustalW as implemented in Unipro ugene v.1.31.0 (Okonechnikov *et al.*, 2012). When necessary, alignments were edited manually. jModeltest2 (Darriba *et al.*, 2012) was used to estimate the best-fit nucleotide substitution model, resulting in selection of HKY + G out of 88 tested models according to the Bayesian information criterion. Phylogenetic analysis was performed in MrBayes v.3.2.6 (Ronquist & Huelsenbeck, 2003) by running two parallel Metropolis-coupled Markov chain Monte Carlo analyses with four chains each for 2 million generations. Tree sampling was conducted every 500 generations. The consensus tree was generated after removal of 25% of samples from the beginning of the chain as burn-in. The final tree was visualized with FigTree v.1.4.3 (http://tree.bio.ed.ac.uk).

### Flow sorting of sperm and vegetative nuclei

Mature pollen grains from anthers at the dehiscence stage were collected and the corresponding nuclei were isolated by applying the filter bursting method (Kron & Husband, 2012) using CellTrics disposable filters of 100 and 20 µm (Sysmex, Norderstedt, Germany). The resulting nuclei suspension was stained with propidium iodide (PI, 50 µg $ml^{-1}$) and analysed on a BD Influx (Becton Dickinson, Franklin Lakes, NJ, USA) by blotting the PI fluorescence against the forward scatter signal (FSC). Based on differences in the DNA content and nuclear size, populations of vegetative and sperm nuclei without and with Bs could be differentiated. The individual fractions were flow-sorted into Eppendorf tubes and fixed with 4% formaldehyde in 5% sucrose buffer (10 mM Tris, 50 mM KCl, 2 mM $MgCl_2$, 5% sucrose, 0.05% Tween, pH 7.5) for 10 min. Subsequently,

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15–20 µl of the nuclei suspensions were applied on microscopic slides and dried overnight before performing immunostaining and fluorescence in situ hybridization (FISH).

### Preparation of tissue sections and chromosome spreads

For the preparation of tissue sections, postmeiotic anthers (5–5.5 mm long) were fixed in 4% paraformaldehyde prepared in 1 × microtubule stabilization buffer (MTSB) as described in Houben *et al.* (2011). In brief, for embedding, the anthers were infiltrated first with a graded series of polyethylene glycol (PEG)-wax in absolute ethanol solutions (1 : 2, 1 : 1, 2 : 1 (v/v)), for 12 h each at 37°C. Finally, anthers were infiltrated with pure PEG-wax for 12 h at 37°C and transferred into casting moulds. Tissue sections (10 µm) were cut on histological microtome Leica RM 2265 (Leica Biosystems) and placed on poly-L-lysine-coated microscopic slides. The preparation of meiotic chromosomes was performed according Badaeva *et al.* (2017). Meiotic anthers (*c.* 1.5 mm long) were fixed in 3 : 1 (ethanol : acetic acid, v/v) for 1 h. Meiocytes were squeezed out of the anthers on a slide in 7 µl of 45% acetic acid with the help of a needle and squashed between slide and coverslip. The coverslips of liquid nitrogen-frozen slides were removed and slides were kept in 1 × PBS at 4°C for immunostaining and FISH. The preparation of mitotic chromosomes from the apical meristem of 3-d-old seedlings was performed according to Aliyeva-Schnorr *et al.* (2015) by dropping the cell suspension on the slides. The slides were stored until use in 96% ethanol at 4°C.

### Immunostaining and FISH

Slides carrying the anther tissue sections were washed in 96% ethanol twice for 10 min, in 90% ethanol twice for 10 min, and finally in $dH_2O$ twice for 5 min. After an 800 W microwave treatment of the slides for 60 s (Chelysheva *et al.*, 2010) in 10 mM sodium citrate buffer, slides were immediately transferred into 1 × MTSB for 5 min. Mouse anti-α tubulin (catalogue no. T 9026, diluted 1 : 200; Sigma), rabbit anti-grass CENH3 (Sanei *et al.*, 2011) (diluted 1 : 3000) and rabbit anti-histone H3K27me3 (#07-449, diluted 1 : 100; Millipore) were used as primary antibodies. Anti-mouse Alexa 488 (catalogue no. A11001, diluted 1 : 200; Molecular Probes, Eugene, OR, USA) and anti-rabbit Cy3 (catalogue no. 111-165-144, diluted 1 : 300; Jackson/Dianova) were used as secondary antibodies. To combine FISH and immunostaining, slides were rinsed with $dH_2O$ and placed in 2 × saline sodium citrate buffer (SSC) for 5 min before FISH. FISH was performed after postfixation of specimens in 3 : 1 fixative for 10 min, washed twice in 2 × SSC and with final fixation in 3% paraformaldehyde. Hybridization mixture (50 µl per slide) was used and the denaturing of tissue sections was done for 5 min on a hot plate at 80°C. For multiple rounds of FISH, FISH signal stripping was performed as described by Heslop-Harrison *et al.* (1992) and Shibata *et al.* (2009). In brief, after removing the coverslips, the slides were rinsed twice in 0.1 × SSC for 5 min at room temperature. Next, slides were incubated in shaking jars at 42°C for 30 min in each solution: once in 0.1 × SSC, twice in probe stripping solution (0.05% (v/v) Tween-20, 50% (v/v) formamide in 0.1 × SSC) and once in 4T solution (0.05% (v/v) Tween-20 in 4 × SSC). Finally, specimens were fixed in 4% formaldehyde for 10 min, dehydrated in an ethanol series and air-dried.

### Generation of FISH probes

The wheat (peri)centromere-specific probes 192 bp (Ito *et al.*, 2004), 6C6-3&4 (Zhang *et al.*, 2004), CCS1 (Aragón-Alcaide *et al.*, 1996), centromere satellite repeat (CenSR) (Li *et al.*, 2013), CRW2 (Liu *et al.*, 2008), pBs301 (Cheng & Murata, 2003), RCS2 (Liu *et al.*, 2008), Quinta-LTR (Li *et al.*, 2013), TaiII (Kishii *et al.*, 2001) and Weg 1-LTR (Li *et al.*, 2013) were generated by PCR (Table S1). Nine of the 11 probes are part of a (peri)centromeric Ty3/gypsy-type retrotransposon. The corresponding positions along the retrotransposon are shown in Fig. S1. Plant genomic DNA was extracted by the CTAB method (Murray & Thompson, 1980). The 25 µl PCR reactions contained the following components: 2.5 µl 10 × PCR buffer (20 mM $MgCl_2$), 1.5 µl dNTP mix (10 mM), 1 µl of each primer (10 mM), 0.2 µl *Taq* (1U) polymerase (Qiagen), 50 ng gDNA and $dH_2O$. Purified PCR products as well as plasmid clones of the newly identified repeats, the rye B-specific repeat D1100 (Sandery *et al.*, 1990) and 5S rDNA clone pTa794 (Gerlach & Dyer, 1980) were directly labelled with dUTP-ATTO-647, dUTP-ATTO-550 or dUTP-ATTO-488 using a nick translation labelling kit (Jena Bioscience, Jena, Germany).

### Microscopy and image analysis

The specimens were analysed using an Olympus BX61 microscope equipped with an ORCA-ER charge-coupled device camera (Hamamatsu, Japan). Deconvolution microscopy was employed to remove out-of-focus information. Maximum intensity projections were processed with the software CellSens Dimension (Olympus). All images were collected in greyscale and pseudo-coloured with Adobe Photoshop CS (Adobe). The ImageJ (https://imagej.nih.gov/ij/download.html) particle analysis tool was used to measure the size of centromere-specific histone variant H3 (CENH3) signals. Graphs showing the CENH3 signal size and the length of spindle microtubules were prepared with the Rstudio ggplot2 package.

## Results

### Identification of a conserved B-specific repeat suitable to trace B chromosomes in *A. speltoides*, *A. mutica* and rye

To find a B-specific marker suitable to track segregation of *A. speltoides* Bs during microgametogenesis by FISH, genomic 0B and +B DNA were paired-end Illumina-sequenced and the corresponding reads were characterized by comparative similarity-based clustering using the RepeatExplorer pipeline (Novák *et al.*, 2013). Highly and moderately abundant repeats represented by clusters containing at least 0.01% of analysed reads



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were found to make up *c.* 77% of the genome. They were mostly composed of LTR-retrotransposons, transposons and satellite repeats, with overall proportions of individual repeat types being similar in the 0B/+B genotypes (Table S3). Comparative cluster analysis revealed four tandem repeats with monomers ranging from 86 to 1090 bp, which were found to be potentially enriched on B chromosomes (Table 1; Fig. 1a,b). The B-specific candidate repeats were PCR-amplified from the +B genotype, cloned and verified by sequencing. In contrast to AesTR-183 and AesTR-205, AesTR-126 and AesTR-148 could also be amplified by using the 0B genotype DNA as template (Fig. S2). FISH with cloned probes resulted in an exclusively B-specific localization of AesTR-183 and AesTR-205, while AesTR-126 and AesTR-148 signals were present on A and B chromosomes (Fig. 1c,d). In all analysed populations, a B-specific distribution of AesTR-183 hybridization signals was found along the long B chromosome arm, whereas AesTR-205 clusters occur on both arms (Fig. 1d). *A. speltoides* from the Katzir population possess, in addition, a morphological variant of the B, presumably lacking the terminal region of the long arm (Ruban *et al.*, 2014). In this metacentric B-variant, AesTR-183 repeats are present on both arms (Fig. S3).

Notably, the tandem repeat AesTR-183 with a monomer length of 1090 bp also displayed B-specific signals in the closely related species *A. mutica* (Fig. 1e). In +B rye, AesTR-183 signals are localized in the nondisjunction-controlling region of the B chromosome (Fig. 1f). To test the presence of this repeat on the Bs of a more distantly related grass species, the AesTR-183 probe was hybridized to a *L. perenne* accession carrying two Bs. However, no B-specific signal was found (data not shown). Hence, three species from the Triticeae encode a conserved B-specific repeat that can be used for the identification of Bs. BLAST (Altschul *et al.*, 1990) search revealed 82% similarity between a part of AesTR-183 and a 489-bp-long region on chromosome 1B of *T. aestivum* (accession LS992081.1). To estimate the relationships of corresponding sequences, a phylogenetic tree was constructed based on sequences of PCR-amplified AesTR-183 monomers from *A. speltoides*, *A. mutica* and rye, including the similar sequence from wheat chromosome 1B (Fig. S4). The obtained phylogenetic tree placed the analysed sequences in three clades according to the species of origin. The much longer branch lengths in the case of rye-derived sequences indicate higher substitution rate. Surprisingly, the sequence from wheat chromosome 1B appeared as a sister group to the rye B-located sequences, whereas it would be expected to cluster together with *A. speltoides*-derived sequences as a result of closer evolutionary relationships of wheat and *A. speltoides* genomes. Interestingly, the diversity of AesTR-183 repeat monomers in *A. mutica* was much lower than in *A. speltoides* and rye.

**Table 1** Repeats significantly enriched on B chromosomes of *Aegilops speltoides*.

| Repeat | Type | Monomer (bp) | Cluster | Reads | | Genome % | |
|---|---|---|---|---|---|---|---|
| | | | | 0B | +B | 0B | +B |
| AesTR-126 | Tandem | 86 | 126 | 1688 | 3528 | 0.123 | 0.258 |
| AesTR-148 | Tandem | 791 | 148 | 934 | 1943 | 0.068 | 0.142 |
| AesTR-183 | Tandem | 1090 | 183 | 1 | 730 | 0.000 | 0.053 |
| AesTR-205 | Tandem | 270 | 205 | 0 | 358 | 0.000 | 0.026 |

## Flow cytometry enables the accumulation of B chromosomes in sperm nuclei to be determined

To quantify the accumulation of *A. speltodies* B chromosomes during pollen development, we applied a combination of flow cytometry, immunostaining and FISH. Differences in size and shape of nuclei isolated from mature pollen grains (Fig. 2a) of a 0B plant allowed the separation of vegetative and sperm nuclei by plotting the fluorescence intensity (DNA content) against the FSC representing particle size. The round-shaped and larger vegetative nuclei (veg) appear in the upper fraction, and the spindle-shaped smaller sperm nuclei were found in the lower fraction (Fig. 2b). To confirm the flow-cytometric separation of both types of nuclei, we sorted both fractions and used the nuclei for immunolabelling with an antibody against histone H3K27me3, which was shown to be preferentially enriched in vegetative nuclei in rye (Houben *et al.*, 2011). Indeed, an intense immunolabelling of nuclei was only observed on samples derived from the upper fraction representing the vegetative nuclei (Fig. 2c), but not on sperm nuclei (Fig. 2d). Thus, both types of nuclei can easily be differentiated by flow cytometry.

In nuclei suspensions of pollen derived from plants with Bs, additional nuclei populations were observed in the dot plots (Figs 2e, S5b–g). Depending on the number of Bs in the mother plant, we detected one to three additional nuclei populations at similar FSC intensities as detected for the sperm nuclei of the 0B plants (Fig. S5a–h) but with a higher fluorescence intensity indicating an increased DNA content as a result of the presence of additional B chromosomes. The number of Bs represented by the individual populations could be estimated by the distance between the fractions of vegetative nuclei, which appeared unchanged, and the corresponding fraction of sperm nuclei. In the case of a 1B plant, one additional population of sperm nuclei appeared at a position that indicates the presence of two B chromosomes. However, the majority of sperms showed no Bs (Fig. S5b). This observation can be explained by the fact that the B chromosome appears as a univalent during metaphase I, leading either to laggards mostly being excluded from the new daughter nuclei or to the migration of the single B chromosome to one of the two poles. In both cases, the second meiotic division occurs normally, but in the latter case two of the four spores contain one B chromosome each (Mendelson & Zohary, 1972) (Fig. 6). Consequently, sperm nuclei formed after two rounds of mitotic divisions contain no or two B chromosomes. In our experiments, we detected 25% of sperm nuclei containing two Bs (Table 2). In pollen of 2B plants, almost exclusively sperm nuclei with two Bs were detected (Fig. S5c). In this case, the two Bs frequently form bivalents during meiosis I and segregate normally. At the end of meiosis, spores with one B chromosome each are formed (Mendelson & Zohary, 1972), which result in sperm nuclei with two chromosomes (Fig. S6). 3B plants produced pollen with more or less equally sized fractions of sperm nuclei with two and four B

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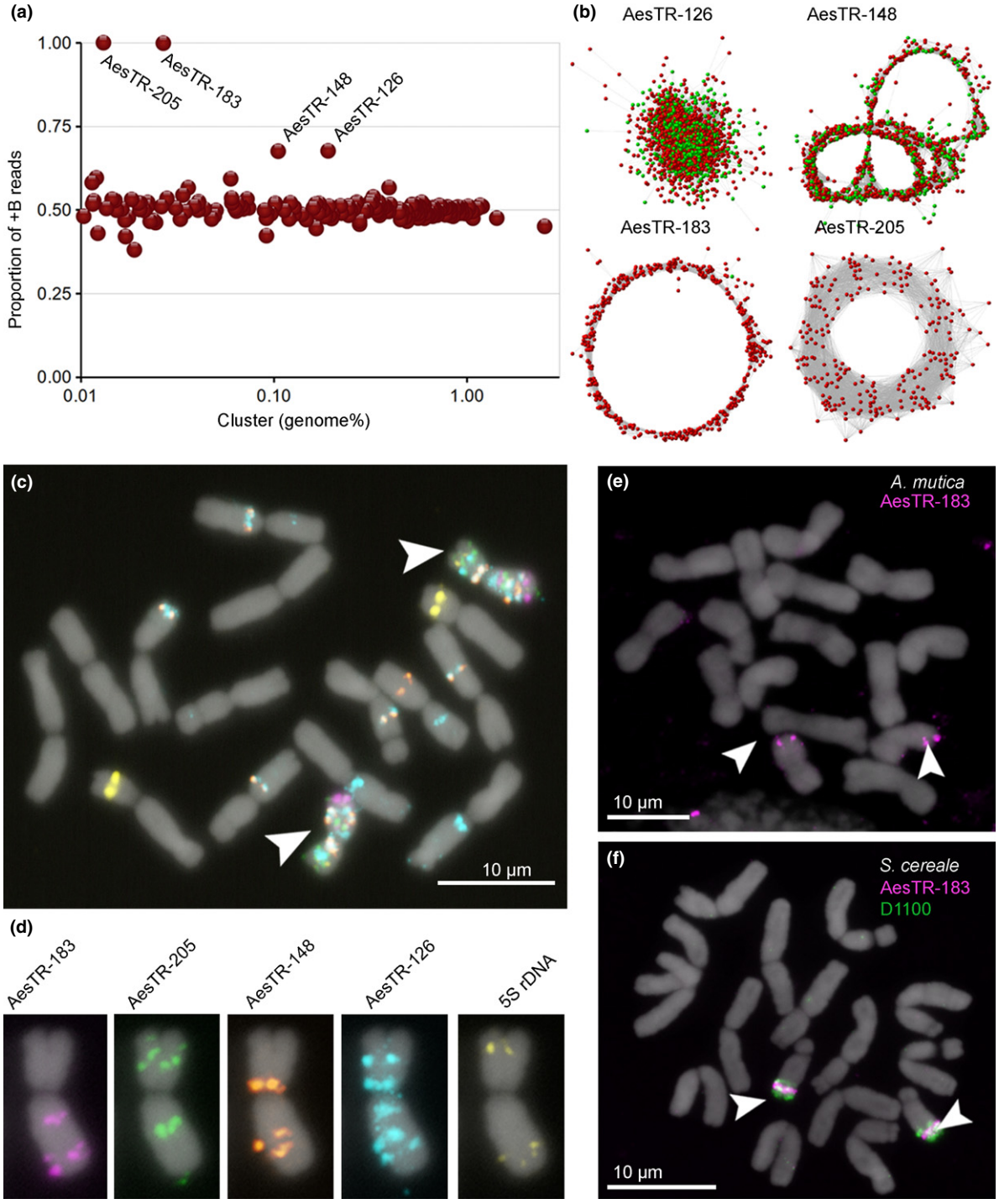



**Fig. 1** Identification and characterization of *Aegilops speltoides* B chromosome (B)-enriched sequences. (a) *In silico* identification of repeat clusters containing B-enriched sequences. The enrichment is revealed by calculating the ratio of reads derived from the +B genotype to the total number of reads (+B and 0B) in the cluster. Clusters showing partial enrichment (AesTR-126 and 148) or specificity to B chromosomes (AesTR-183 and 205) are labelled. (b) Graph representation of read similarities in B-enriched repeat clusters. Individual reads from 0B and +B plants are represented by green and red nodes, respectively. Nodes are placed according to sequence similarity, where similar sequences are close together and connected with edges (grey lines). The circular or globular structure of these graphs is typical for repeats with tandem arrangement in the genome. (c–f) Chromosomal distribution of B-enriched repeats in *A. speltoides*, *Aegilops mutica* and *Secale cereale*. (c) Localization of AesTR-126, AesTR-148, AesTR-183, AesTR-205 and 5S rDNA on mitotic metaphase chromosomes of *A. speltoides* from Tartus (Syria) revealed by fluorescence *in situ* hybridization (FISH). (d) Individual hybridization patterns of FISH probes. (e) Colocalization of AesTR-183 (magenta) and the *S. cereale* B-specific probe D1100 (in green) in *S. cereale* with 2Bs. (f) Localization of AesTR-183 in *A. mutica* with two Bs. The Bs are marked with arrowheads.



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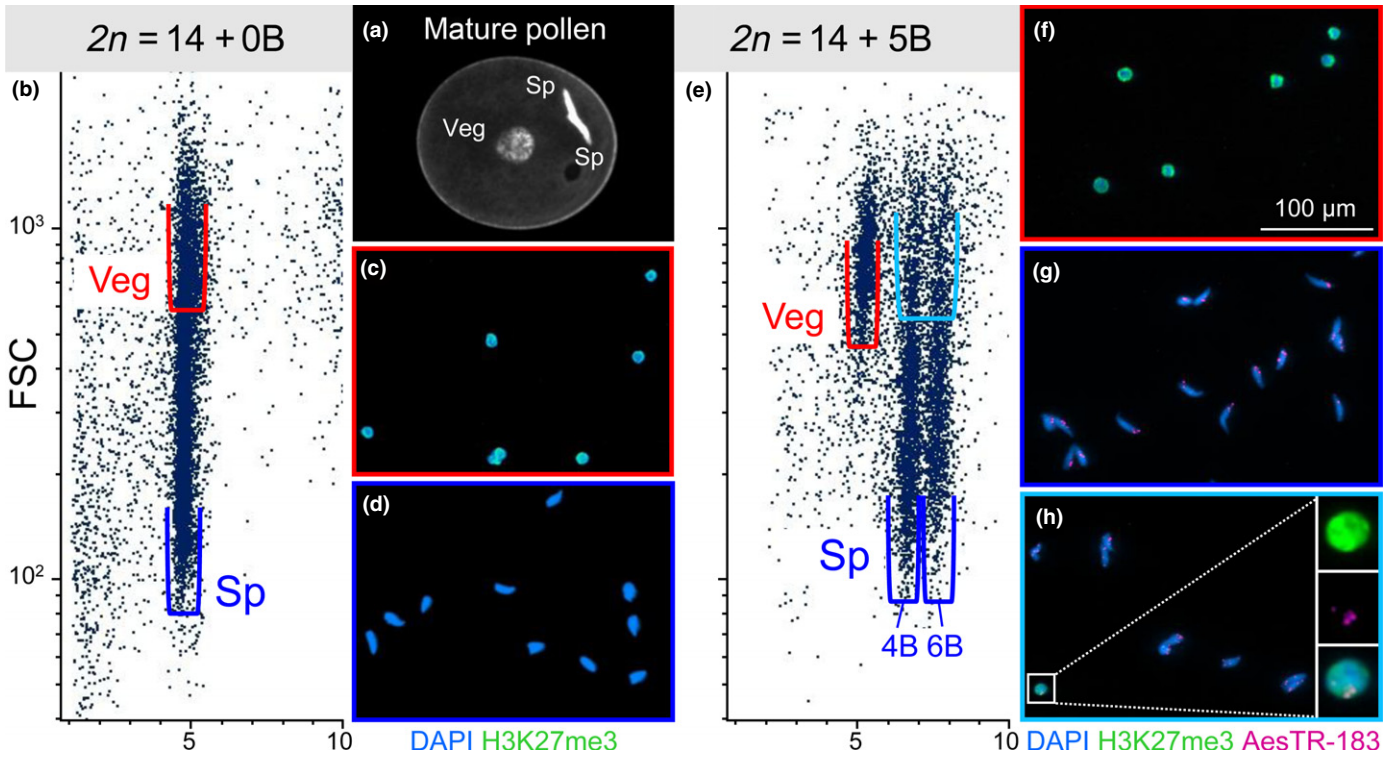



**Fig. 2** Flow cytometry revealed the high efficiency of *Aegilops speltoides* B chromosome drive. (a–d) Vegetative (Veg, red) and sperm (Sp, blue) nuclei of 0B mature pollen were separated by flow cytometry, and the purity of both fractions was verified by H3K27me3 immunostaining. (e, f) Vegetative (red) and sperm (blue) nuclei of +5B mature pollen were sorted and subjected to H3K27me3 immunostaining and B-specific fluorescence *in situ* hybridization (FISH) using AesTR-183. (a) Propidium iodide (PI)-stained mature pollen grain from an 0B plant contained round vegetative nucleus and two spindle-shaped sperm nuclei. (b) Dot plot displaying the relative fluorescence (DNA content; *x*-axis) vs forward scatter signal (particle size; *y*-axis) intensities. (c) The sorted vegetative nuclei fraction (red) was abundantly labelled by anti-H3K27me3 (in green). (d) Sperm nuclei fraction (blue) showed no anti-H3K27me3 signal. (e) Dot plot of +5B plant pollen with fraction of vegetative (red) and two fractions of different fluorescence intensities of sperm nuclei (blue). (f) Sorted vegetative nuclei labelled by H3K27me3 showed no B-specific FISH signals. (g, h) Sorted sperm nuclei after immunostaining with H3K27me3 and FISH with the B-specific probe AesTR-183. Whereas in the lower part (blue) of the sperm nuclei fractions, exclusively B-carrying sperm nuclei (g) were detected, the upper fraction (light blue) also contained a low percentage of vegetative nuclei with B chromosomes (h).

chromosomes (Fig. S5d), similar to 5B plants, where nuclei with four and six B chromosomes (Fig. S5e) were detectable. For 3B plants, the occurrence of univalents, bivalents and trivalents was described (Mendelson & Zohary, 1972). Resolving these pairing structures preferentially results in spores with either one or two B chromosomes finally leading to sperm nuclei with two or four Bs (Fig. S6). However, with a much lower frequency, sperm nuclei with no and six Bs could also be detected (Fig. S5d). In 5B plants, the number of potential pairing configurations increases further. However, preferably either two or three Bs migrate to opposite poles during meiosis I, finally resulting in sperm nuclei with either four or six Bs. In pollen of plants with four and six Bs preferentially sperm nuclei with the same number of B chromosomes as in the corresponding mother plant were produced and with a lower frequency of nuclei with two and six or four and eight Bs, respectively (Fig. S5f,g). Plants with four or six Bs preferentially perform regular segregations during meiosis independent of the pairing configuration (bivalents up to hexavalents) (Mendelson & Zohary, 1972). Deviations from this normal segregation may lead to a decrease or increase in the number of B chromosomes in sperm nuclei compared with the situation in the mother plant (Table 2).

To measure the efficiency of the B accumulation, vegetative and sperm nuclei from mature pollen of individual +B plants were separately sorted onto microscopic slides first (red and blue frames in Fig. 2e). Application of immunolabelling using anti-histone H3K27me3 to identify the vegetative nuclei in combination with FISH using the B-specific repeat AesTR-183 to score

**Table 2** Quantification of *Aegilops speltoides* B chromosome (Bs) accumulation in sperm nuclei.

| Mother plant | Drive (%)[1] (number of nuclei counted) | Frequency of Bs in sperm nuclei (%)[2] | | | | |
|---|---|---|---|---|---|---|
| | | 0B | 2B | 4B | 6B | 8B |
| 1B | 94.4 (407) | 75.1 | 24.9 | | | |
| 2B | 93.9 (770) | 1.9 | 98.1 | | | |
| 3B | 98.3 (754) | 6.5 | 48.0 | 39.5 | 5.9 | |
| 4B | 97.9 (660) | 2.1 | 14.4 | 74.2 | 9.2 | |
| 5B | 96.3 (589) | 1.4 | | 57.3 | 41.4 | |
| 6B | nd | 2.5 | | 24.6 | 61.7 | 11.2 |

[1]Bs drive efficiency calculated based on vegetative nuclei without Bs after flow sorting, immunostaining and fluorescence *in situ* hybridization (FISH). nd, no data.
[2]Frequencies were quantified by flow cytometry.

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for the presence of Bs (Fig. 2f,g) confirmed the purity of the individual populations. In order to evaluate if the B chromosome-containing fractions with a FSC signal intensity similar to that of the vegetative nuclei (light blue frame in Figs 2e, S5b–h) contain sperm nuclei exclusively or whether they also contain vegetative nuclei with Bs, we sorted the upper (light blue) and lower (dark blue) fractions of sperm nuclei separately. Only the upper sperm nuclei fractions (light blue) contained, in addition, a low percentage of vegetative nuclei harbouring Bs (Fig. 2h) in the range 1.7–6.1%. Based on these cytological evaluations in combination with the flow cytometric profiles, we were able to precisely determine the transmission of Bs to sperm nuclei (Table 2). Independent of the number of Bs present in the mother plant, Bs accumulate in the sperm nuclei to >93% as a result of an unequal segregation of Bs during the first pollen mitosis.

Because in rye a structural change of the B chromosome impaired its ability to drive (Müntzing, 1945; Endo *et al.*, 2008), we included *A. speltoides* plants from the Katzir population with a metacentric B-variant in our analysis (Fig. S3b). However, based on flow cytometry, this B-variant shows a comparable frequency of accumulation in sperm nuclei (Fig. S5h). Hence, the changes involved in the formation of this B-variant did not affect the process behind the control of its preferential transmission to sperm nuclei.

## B chromosomes preferentially distribute in the generative nucleus of the bicellular pollen grain

To obtain visual insights into the cellular mechanism of B chromosome drive in *A. speltoides*, we analysed the dynamics of chromosomes at microgametogenesis. Anther tissue sections were used to maintain the natural arrangement of chromosomes and to avoid pollen wall background signals. The B-specific probe AesTR-183 was used to trace the segregation of Bs during pollen formation.

Products of the pollen mitosis I were analysed first. Owing to differences in the chromatin condensation and size of generative and vegetative nuclei (Tanaka, 1997; Borg *et al.*, 2009), both types of nuclei are easily distinguishable. Condensed generative nuclei locate close to the pollen wall. In all, 120 bicellular pollen grains of plants possessing 2Bs were analysed. In 85% of pollen, the Bs were observed in the generative nucleus only (Fig. 3a). Micronuclei containing Bs were found in close vicinity to the

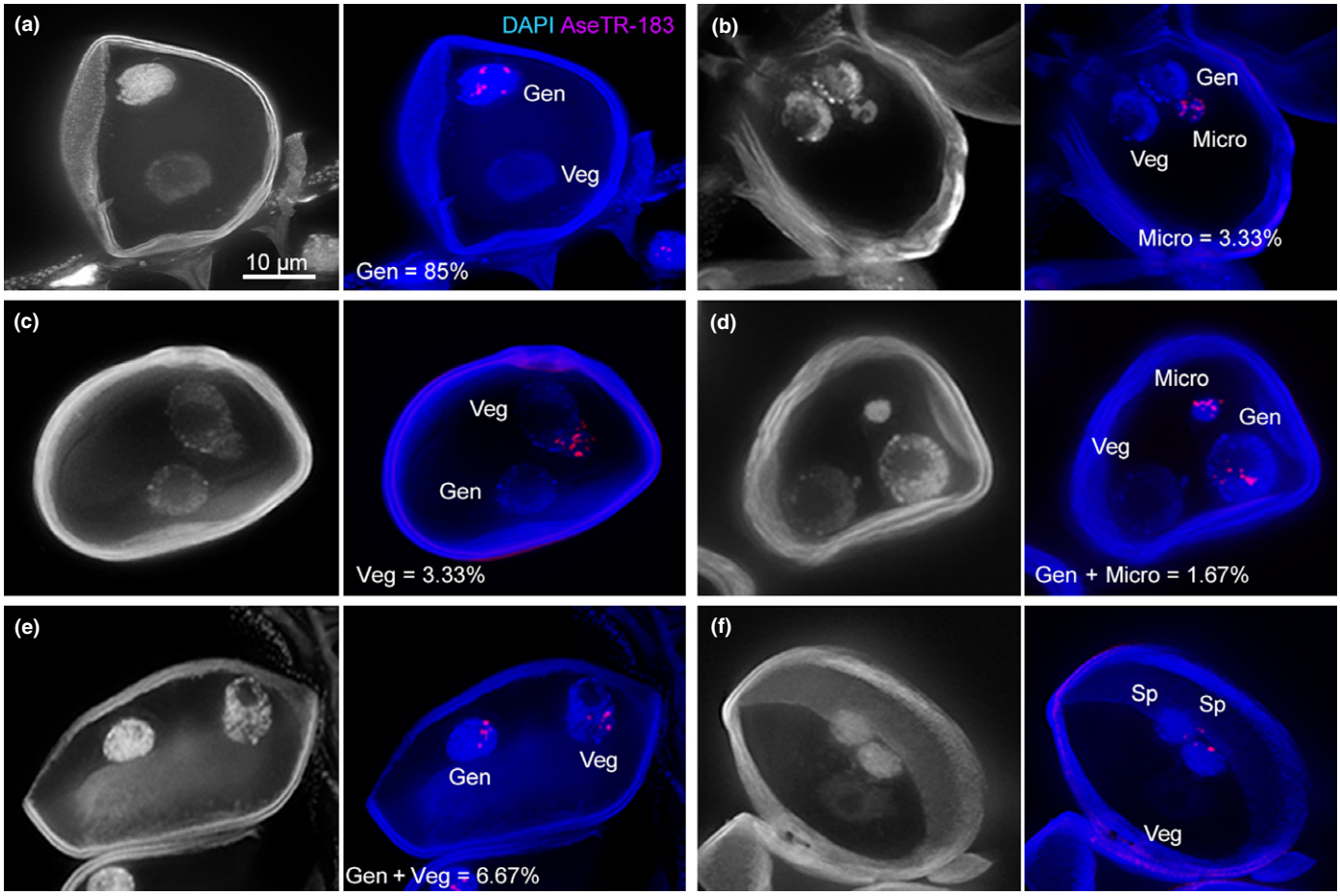



**Fig. 3** B chromosomes (Bs) of *Aegilops speltoides* preferentially accumulate in the generative nucleus during first pollen grain mitosis. Tissue sections after fluorescence *in situ* hybridization (FISH) with the B-specific repeat AesTR-183 (magenta): (a) 85% of binucleated pollen possess Bs in generative nuclei (Gen) only; (b) 3.33% of binucleated pollen possess Bs in micronuclei (Micro) only; (c) 3.33% of binucleated pollen possess Bs in vegetative nuclei (Veg); (d) 1.67% of binucleated pollen possess Bs in generative and micronuclei; (e) 6.67% of binucleated pollen possess Bs in generative and vegetative nuclei; (f) in mature pollen, both sperm nuclei possess Bs. 4′,6-Diamidino-2-phenylindole (DAPI) is shown in blue.



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generative nucleus in 3.3% of pollen (Fig. 3b); 3.3% of pollen revealed B-specific signals in vegetative nuclei only (Fig. 3c) and 1.7% of pollen revealed Bs in the generative nucleus and in micronuclei (Fig. 3d); 6.7% of pollen showed B-specific signals in both generative and vegetative nuclei (Fig. 3e). In all 10 analysed mature tricellular pollen grains, B-specific signals were never observed in vegetative nuclei, while all of the sperm nuclei revealed signals, indicating a normal segregation of Bs at second pollen grain mitosis (Fig. 3f). Thus, Bs accumulate in the generative nucleus during the first pollen mitosis. The observed values of the B-chromosome accumulation in tissue sections are in close agreement with the flow cytometric data.

### Directed nondisjunction of *A. speltoides* Bs is accompanied by unequal cell division during pollen mitosis

To decipher the mechanism behind the preferential accumulation of B chromosomes in the generative nucleus, tubulin and CENH3 antibodies were applied at all stages of the first pollen mitosis of B chromosome-containing plants. The centromere-specific histone variant H3 (CENH3) was selected because it was shown that its loss results in the failure of chromosome segregation in mammals (Howman *et al.*, 2000) and plants (Sanei *et al.*, 2011). The centromeres of all anaphase A and B chromosomes were equally CENH3-labelled and revealed an interaction with microtubules. During early anaphase, the peripheral generative spindle bundle is blunt and the pole is nearly in contact with the pollen cortex. By contrast, the interior spindle bundle is long and pointed (Fig. 4a). In late anaphase, a nondisjoined lagging chromosome with attached spindle fibres was observed (Fig. 4b). During telophase, the generative nucleus contained a higher number of CENH3 signals indicating the presence of Bs (Fig. 4c). Comparison of the distance between the spindle midzone and the centromere positions of the generative (6.82 μm) vs the vegetative nucleus (7.97 μm) revealed that, in most cases, the spindle midzone is more proximal to the generative nucleus ($n = 10$; *t*-test, $P < 0.05$) (Fig. 4d).

In order to quantify the centromere size as a means for centromere activity (Zhang & Dawe, 2012), the size of immunosignals was determined after anti-CENH3 labelling and B-specific *in situ* hybridizations of somatic mitotic +B metaphase cells. No major differences in CENH3 signal size were found when we compared A and B centromeres, although the amount of CENH3 varied slightly between the individual chromosomes (Fig. 5a,b). Similarly, we did not observe differences in CENH3 signal size using meiotic prophase cells. Thus, a different centromere activity of B chromosomes could be excluded as the reason for the nondisjunction of Bs, as the centromeres of all chromosomes are CENH3-positive and interact with tubulin.

### The (peri)centromere composition differs between A and B chromosomes of *A. speltoides*

Next, we asked whether the sequence composition differs between A and B (peri)centromeres. Eleven (peri)centromere-specific repeats, originally described for wheat, were identified in *A. speltoides*. The corresponding FISH probes were generated by PCR using genomic DNA of *A. speltoides* carrying B chromosomes (Table S1) and hybridized to mitotic metaphase chromosomes of +B plants (Figs 6, S7; Table 3). Four of the 11 probes (RCS2, 6C6-3, 6C6-4 and CRW2) resulted in signals of similar intensities on A and B chromosomes (Figs 6a, S7a–c). Four probes (Quinta-LTR, 192 bp, Weg 1-LTR and CCS1) showed stronger hybridization signals on the A chromosomes (Figs 6b, S7d–f), and pBs301 and the centromeric satellite repeat (CenSR) colocalized with the pericentromeres of As but were not detectable on Bs (Figs 6c, S7g). By contrast, TaiII revealed an enriched signal accumulation at centromeres of Bs (Fig. 6d). No obvious differences in the distribution and intensity of the used (peri)centromeric probes were found between the different A chromosomes of *A. speltoides*.

To evaluate the centromeric localization of the identified repeats in more detail, we used naturally extended pachytene chromosomes for anti-CENH3 staining and subsequent FISH with the corresponding repeats. AesTR-183 was additionally used to distinguish between A and B chromosomes. Quinta-LTR, 192 bp, Weg 1-LTR (Figs 6b′, S7d′,e′) signals partly overlapped with CENH3-marked chromatin and extended into pericentromeric regions. CCS1 that completely overlapped with CENH3 signals in A chromosomes showed reduced centromere labelling in B chromosomes (Fig. S7f′). No overlapping with CENH3-signals was found for the A chromosome-specific pBs301 and CenSR (Figs 6c′, S7g′). TaiII coincided with CENH3 and showed stronger centromere signals on Bs than on As (Fig. 6d′). Although nine of the 11 (peri)centromeric candidate sequences colocalize with CENH3 signals, CENH3 never encompassed fully the area covered by centromere repeats. We conclude that all A centromere sequences except pBs301 and the CenSR are also present in the centromeres of the Bs. Hence, an intraspecific origin of the B centromere is likely. However, the contrasting abundance of Quinta-LTR, 192 bp, Weg 1-LTR and CCS1 signals indicates that the (peri)centromere of the B diversified.

## Discussion

### AesTR-183, a conserved B chromosome-specific repeat

The high copy-repeat fraction of the *A. speltoides* genome is mostly composed of LTR-retrotransposons, transposons and satellite repeats, with overall proportions of individual repeat types being similar in the 0B/+B genotypes. Among four identified B-located repeats in *A. speltoides*, only AesTR-183 and AesTR-126 showed similarity to already known repeats in other species according to BLAST results. AesTR-126 appeared to be identical to a part of the *T. aestivum* clone pTa-713, a repetitive sequence positively tested in FISH on wheat chromosomes (KC290900.1) (Badaeva *et al.*, 2015). AesTR-183 shows 84% similarity to the rye B-specific tandem repeat Sc26c38 which occurs within the nondisjunction control region of the rye B (Klemme *et al.*, 2013). In addition, we discovered an AesTR-183-like repeat in the Bs of *A. mutica*. It is unknown whether the Bs of *A. speltoides* and *A. mutica* possess,

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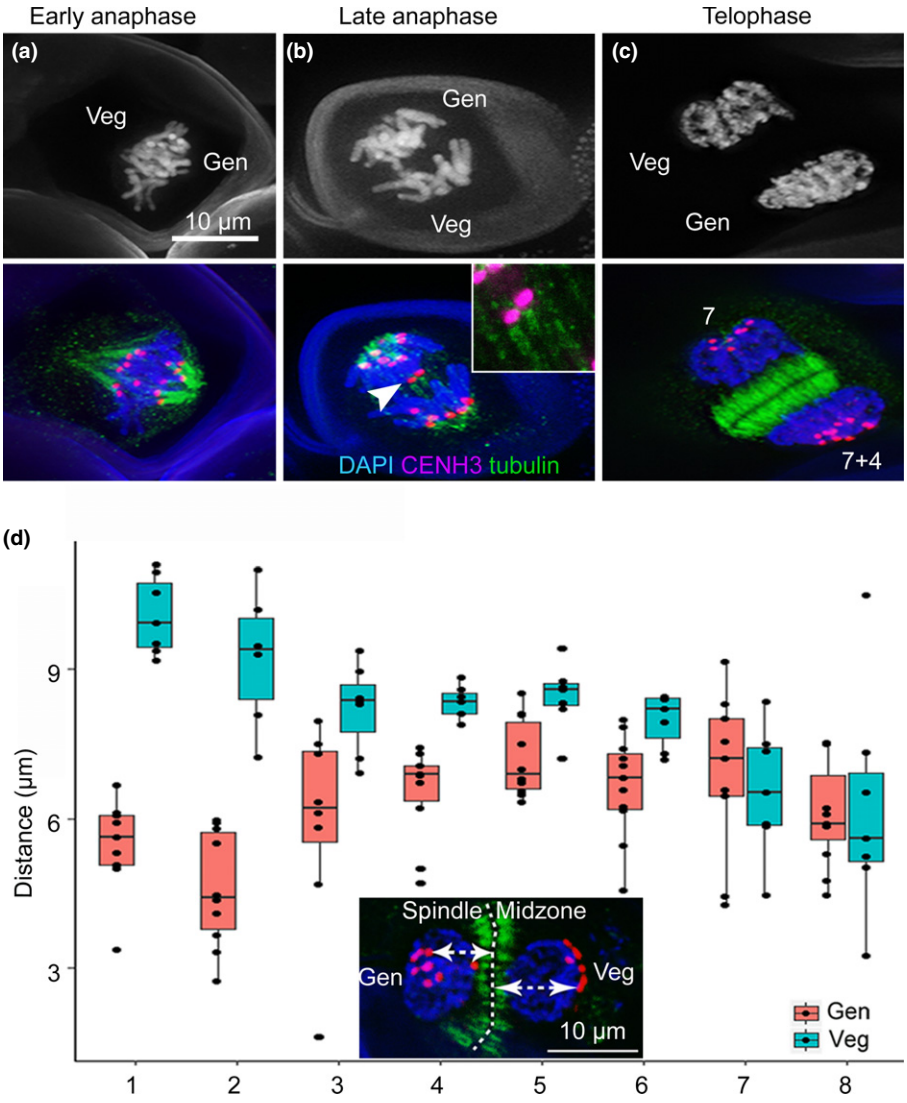



**Fig. 4** The accumulation of B chromosomes (Bs) during pollen mitosis of *Aegilops speltoides* +2B. (a–c) CENH3 (magenta) and α-tubulin (green) immunolabelling patterns at different stages of first pollen mitosis. (a) Formation of asymmetric spindle at early anaphase. The generative spindle (Gen) is short and blunt and the vegetative spindle (Veg) is long and sharp. (b) At late anaphase, nondisjunction of lagging chromosomes with positive CENH3 signals attached to the tubulin (insert shows further enlarged CENH3 signals). (c) Pollen at late anaphase forming an asymmetric spindle midzone. Note the unequal number of CENH3 signals. (d) The dots indicate the distance between the spindle midzone and the individual centromeres of the generative nucleus (red) or vegetative nucleus (blue). Error bars represent the deviation from upper and lower quartile based on measurements conducted on the distance between centromere and spindle midzone. *x*-axis, individual cells; *y*-axis, distance.

like rye, a repeat-enriched nondisjunction control region. The existence of a B chromosome conserved repeat is intriguing and provokes the question of whether the AesTR-183 repeat has acquired a function in Bs. Possibly, the ancestor of all three species possessed an AesTR-183-positive B chromosome, which evolved further in rye and *Aegilops*, but kept the AesTR-183 sequence. The phylogenetic analysis of AesTR-183 repeat monomer sequences tends to support this theory, as the divergence of B-specific repeat in this case coincides with divergence of the species themselves. The similar but much shorter sequence detected on wheat chromosome 1B does not follow this pattern and seems to be independent of the B-specific sequences.

## A combination of nondisjunction and unequal spindle formation at first pollen mitosis might promote the accumulation of Bs in sperm nuclei

We provide direct insight into the cellular basis of the drive mechanism of the B chromosome. The flow-cytometrically estimated frequencies of B chromosome accumulation correspond to the cytologically determined one using tissue sections of developing pollen grains (93.9% vs 95%). However, one advantage of the cytological approach is the observation of micronuclei, which escaped from detection in the flow cytometry approach. On the other hand, flow cytometry allows the rapid analysis of a large number of pollen nuclei of different plants. This approach seems also to be applicable in other plant species with B chromosomes, provided the structural and morphological differences between vegetative and sperm nuclei allow a clear differentiation, and the size of the B chromosomes is large enough to separate individual fractions with different B chromosome numbers.

More than 93% of sperm nuclei accumulated Bs independent of the number of Bs present in the mother plant. In +6B plants, sperm nuclei with up to eight Bs were found. Eight reflects the maximum number of Bs detected in *A. speltoides*. Thus, embryos with a B number > 8 are not formed or do not develop as a result of the negative effect of Bs in high numbers. It would be of interest to investigate pollen of +8B plants to see if at least sperm nuclei with more than eight Bs are produced. However, plants



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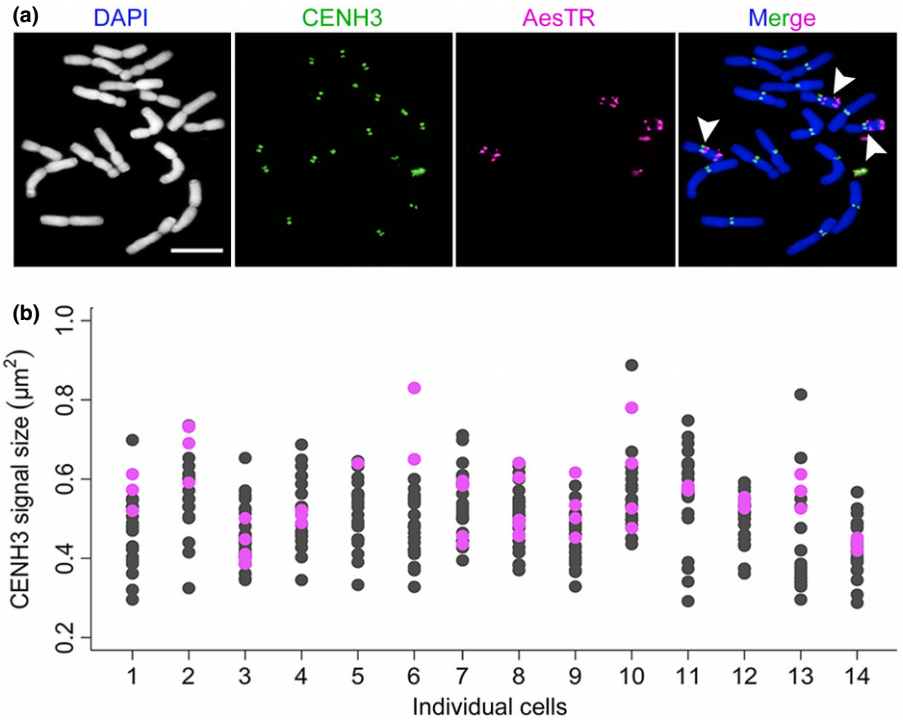



**Fig. 5** The size of CENH3 immunosignals does not differ between mitotic A (As) and B chromosomes (Bs) of *Aegilops speltoides*. (a) Mitotic metaphase of a 3B *A. speltoides* plant after CENH3 immunostaining (green) and fluorescence *in situ* hybridization with the B-specific probe AesTR-183 (magenta). Bs are marked with arrowheads. Bars, 10 μm. (b) Each dot represents the CENH3 signal size of an individual sister chromatid (As in black, Bs in magenta). *X*-axis: individual cells, *Y*-axis: CENH3 signal size.

with eight Bs were not found during this study and are probably male-sterile.

An important hint regarding the mechanism of chromosome accumulation comes from our finding that the microtubule spindle of *A. speltoides* is asymmetrical during first pollen mitosis, in accordance with previous studies in other species (Tanaka, 1997; Jin *et al.*, 2005; Borg *et al.*, 2009; Twell, 2011; Banaei-Moghaddam *et al.*, 2012). Considering the asymmetric geometry of the spindle at first pollen grain mitosis, it is likely that the inclusion of Bs into the generative nucleus is caused by the fact that the equatorial plate is nearer to the generative pole, as suggested by Jones (1991). Spindle asymmetry as a component of the B chromosome drive process has also been suggested for the lily *Lilium callosum* (Kimura & Kayano, 1961), the Asteraceae *Crepis capillaris* (Rutishauser & Rothlisberger, 1966) and the grasshopper *Myrmeleotettix maculatus* (Hewitt, 1976). Alternatively, as a result of a higher pulling force on the B centromere towards the generative pole, Bs may preferentially accumulate in the generative nucleus (Banaei-Moghaddam *et al.*, 2012).

Mitotic nondisjunction and lagging chromatids could be explained by centromere inactivity (Sanei *et al.*, 2011). However, as the centromeres of lagging Bs show no reduction of CENH3 signals and interact with spindle microtubules, centromere inactivity is unlikely to be the reason for the nondisjunction process. In rye, the failure to separate CENH3-positive B sister pericentromeres during pollen grain mitosis is under the control of the nondisjunction control region (Endo *et al.*, 2008; Banaei-Moghaddam *et al.*, 2012). Also in maize, centromere function and B chromosome nondisjunction are two independent processes, but the region required for nondisjunction resides in the centromeric region (Han *et al.*, 2007). It is likely that in *A. speltoides* and rye the cohesion between B sister chromatids during first pollen mitosis is stronger than the microtubule traction force required for separation of chromatids. At second pollen grain mitosis, B sister chromatids disjoin and a normal segregation of Bs occurs.

Despite the independent origin of *A. speltoides* and rye, the ancestors of rye and *A. speltoides* diverged more than 4.1–3.4 Ma (Bernhardt *et al.*, 2017), the B chromosome accumulation mechanism in both species is similar. In both species, nondisjunction of CENH3-positive B sister chromatids and an asymmetric spindle during first pollen grain mitosis are likely components of the accumulation process. Assuming that the Bs of rye and *A. speltoides* originated independently, the comparable drive mechanisms in both species probably evolved in parallel.

## Similarities in the (peri)centromere composition of As and Bs suggest an intraspecific origin of B chromosomes in *A. speltoides*

The observed generally similar sequence composition of the A and B centromeres of *A. speltoides* could be considered as evidence of an intraspecific A chromosome-derived origin of the B chromosome. Nine out of 11 tested (peri)centromeric sequences were shared by both types of chromosomes. Only CenSR and pBs301 did not reveal a hybridization signal in the B centromeres. However, copy number differences assumed by different FISH signal sizes were also found for the repeats Quinta-LTR, 192 bp, Weg 1- LTR, CCS1 and Taill. One might imagine that the B centromere evolved from a standard A chromosome centromere and the repeats rearranged in the newly formed B chromosome. A comparable situation was also observed for the B chromosome

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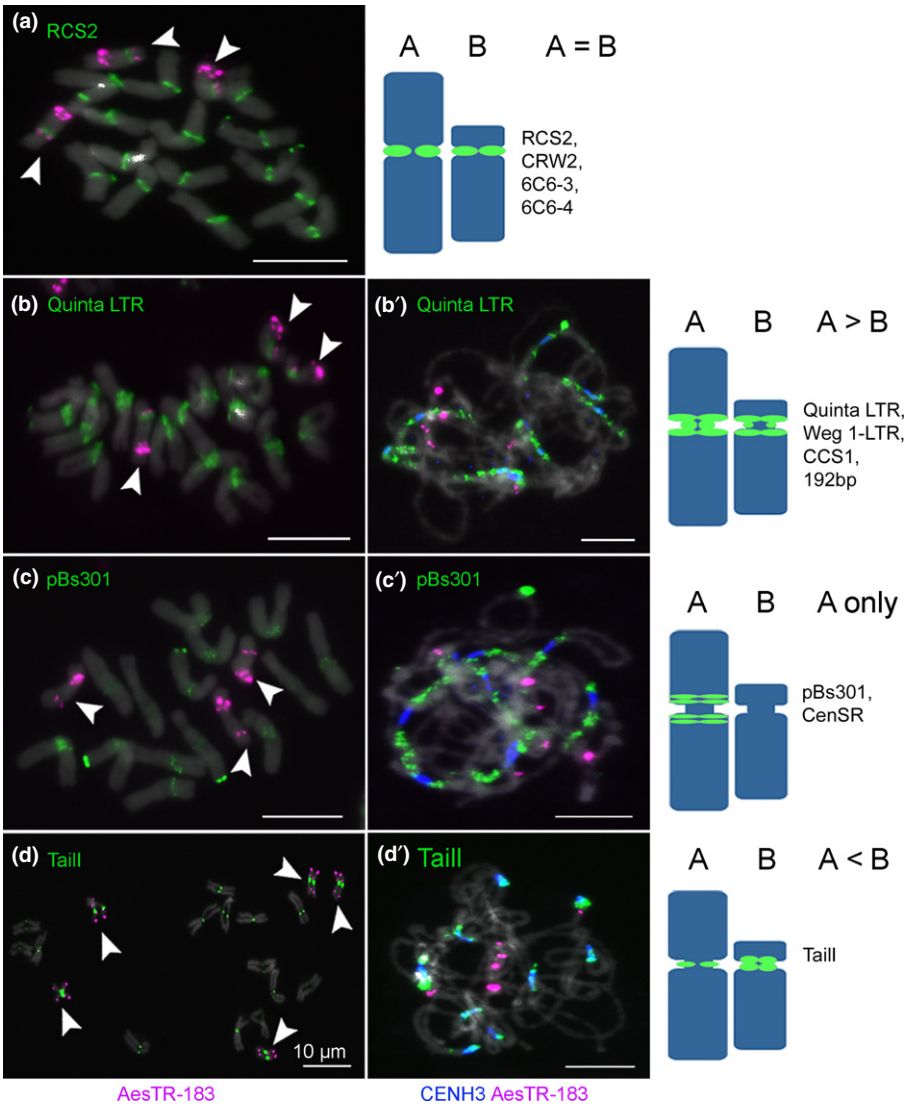



**Fig. 6** Sequence composition of *Aegilops speltoides* A and B (peri)centromeres. (a–d) Fluorescence *in situ* hybridization (FISH) patterns of RCS2, Quinta-LTR, pBs301 and TaiII (green) as well as of the B-specific repeat AesTR-183 (magenta) on mitotic metaphase chromosomes. White arrows indicate B chromosomes (Bs). (b′–d′) Meiotic pachytene chromosomes labelled with the corresponding (peri)centromeric probes (except for RCS2) and with antibodies against CENH3 (blue). Schematic chromosomes represent typical (peri)centromeric distributions of the corresponding sequences on A and B chromosomes. Hybridization patterns of the remaining (peri)centromeric sequences are shown in Supporting Information Fig. S6. 4′,6-Diamidino-2-phenylindole staining is shown in grey. Bars, 10 µm.

**Table 3** (Peri)centromeric fluorescence *in situ* hybridization (FISH) probes.

| | Signal intensity and size[1] | | CENH3 interaction[2] | | |
|---|---|---|---|---|---|
| (peri)centromeric repeat | As | Bs | ChIP (original paper) | FISH in pachytene of *Aegilops speltoides* | Sequence type |
| 192 bp (Ito *et al.*, 2004) | ** | * | + | + | Ty3/gypsy LTR |
| 6C6-3 (Zhang *et al.*, 2004) | *** | *** | + | + | Ty3/gypsy cereba gag |
| 6C6-4 (Zhang *et al.*, 2004) | *** | *** | + | + | Ty3/gypsy cereba gag |
| CCS1 (Aragón-Alcaide *et al.*, 1996) | *** | * | + | + | Ty3/gypsy LTRs |
| Centromeric satellite repeats (CenSR) (Li *et al.*, 2013) | ** | – | – | – | Centromeric satellite |
| CRW2 (Liu *et al.*, 2008) | *** | *** | + | + | Ty3/gypsy cereba LTR |
| pBs301 (Cheng & Murata, 2003) | ** | – | + | – | Ty3/gypsy family |
| RCS2 (Liu *et al.*, 2008) | *** | *** | + | – | Ty3/gypsy family |
| Quinta-LTR (Li *et al.*, 2013) | ** | * | + | + | Ty3/gypsy Quinta LTR |
| Tail-family (Kishii *et al.*, 2001) | * | *** | nd | + | Tandem repeat |
| Weg 1-LTR (Li *et al.*, 2013) | ** | * | + | + | Ty3/gypsy Weg 1-LTR |

[1]*, weak signal; **, medium strong signal; ***, strong signal; –, no signal detected.
[2]+, (peri)centromere colocalized with CENH3 antibody; –, no interaction; nd, no data available; ChIP, chromatin immunoprecipitation.



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centromeres of the daisy *Brachycome dichromosomatica* (Houben *et al.*, 2001), maize (Jin *et al.*, 2005), rye (Banaei-Moghaddam *et al.*, 2012) and the grasshopper *Xyleus discoideus angulatus* (Bernardino *et al.*, 2017). It is likely that the B-specific centromere composition is functionally involved in the regulation of chromosome segregation to ensure the maintenance of Bs in natural populations.

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The authors declare that they have no competing interests.

## Author contributions

DW, AR, JF, YZ and AH designed the study, interpreted the data and wrote the manuscript. MV, PN, JM performed the bioinformatics analysis. JF conducted the flow cytometry. All authors read and approved the final manuscript.

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## Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

**Fig. S1** Schemata of a Ty3/gypsy retrotransposon showing the positions of the used (peri)centromeric FISH probes.

**Fig. S2** PCR amplification of AesTR-126, AesTR-148, AesTR-183 and AesTR-205 repeats from *A. speltoides* genomic DNA of 0B and +B plants.

**Fig. S3** Distribution of the B-specific probes AesTR-183 (magenta) and AesTR-205 (green) on the two B chromosome variants of *A. speltoides*.

**Fig. S4** Phylogeny of AesTR-183-like sequence.

**Fig. S5** Flow cytometry efficiently showed *A. speltoides* B chromosome accumulation in sperm nuclei of +B plants compared with a 0B plant.

**Fig. S6** Scheme of the B chromosome transmission process in the male gametogenesis of *A. speltoides* exemplified for plants with 1B, 2B and 3B chromosomes.

**Fig. S7** Sequence composition of *A. speltoides* A and B (peri)centromeres.

**Table S1** Consensus monomer sequences of AesTR-126, AesTR-148, AesTR-183 and AesTR-205 determined by TAREAN analysis and used to design the primers.

**Table S2** PCR primers used for amplification of AesTR-126, AesTR-148, AesTR-183 and AesTR-205 repeats, control primers for 18S rDNA and primers for (peri)centromeric sequences.

**Table S3** Repeat composition of *A. speltoides* 0B and +B genotypes estimated from clustering analysis of Illumina reads.

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